

SHOW LOW, AZ, USA

WMO: 723747

Lat: 34.264N

Lon: 110.008W

Elev: 1954

StdP: 79.95

Time Zone: -7.00 (W07)

Period: 05-19

WBAN: 03101

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.5	-10.0	-21.0	0.7	1.1	-18.8	0.9	1.5	15.3	6.1	13.0	5.5	1.3	80	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	32.2	13.5	30.9	13.4	29.0	13.2	17.0	23.7	16.5	23.5	16.0	23.1	4.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	13.8	18.6	14.1	12.8	18.4	13.7	12.4	18.4	56.1	24.4	54.2	23.2	52.3	23.2	20.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
14.1	11.8	10.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-17.6	34.4	3.1	0.9	-19.8	35.1	-21.7	35.7	-23.4	36.2	-25.7	36.9
			-18.1	18.2	2.9	1.0	-20.2	18.9	-21.9	19.4	-23.6	20.0	-25.7	20.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	0.6	3.2	7.0	10.1	14.1	20.7	22.0	20.7	17.8	12.0	6.5	1.0		
	DBStd	8.36	4.44	4.54	3.85	3.77	3.91	2.78	2.09	2.12	2.70	3.80	4.33	4.76		
	HDD10.0	1069	292	192	106	44	12	0	0	0	1	24	119	280		
	HDD18.3	2859	550	424	352	247	138	11	1	4	39	198	356	538		
	CDD10.0	1556	0	1	12	47	138	320	373	332	234	84	13	1		
	CDD18.3	304	0	0	0	0	6	81	116	78	23	1	0	0		
	CDH23.3	3477	0	0	0	9	181	1227	1029	702	307	22	0	0		
	CDH26.7	1084	0	0	0	1	27	454	369	191	42	1	0	0		
(14) Wind	WSAvg	4.0	3.3	4.3	4.6	5.2	4.8	4.6	3.6	3.2	3.6	3.7	3.8	3.8		
(15) Precipitation	PrecAvg	354	28	22	21	15	12	10	56	70	37	30	22	32		
	PrecMax	553	106	67	55	49	105	29	117	138	101	82	65	95		
	PrecMin	198	2	0	0	0	0	0	11	23	2	0	0	0		
	PrecStd	72	29	18	17	13	23	7	33	33	25	21	19	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	17.9	22.0	24.1	28.9	33.0	33.7	32.1	29.1	26.0	22.1	16.3		
		MCWB	4.5	6.1	7.5	8.9	11.5	13.0	13.6	14.6	13.9	11.8	8.6	4.7		
	2%	DB	12.4	17.0	19.7	22.6	27.3	32.3	32.2	30.1	27.7	23.7	19.1	13.7		
		MCWB	3.8	5.3	6.9	8.1	10.4	12.7	14.3	14.4	13.2	10.5	7.3	4.3		
	5%	DB	10.2	13.9	17.7	21.2	25.8	31.1	30.8	28.8	27.1	22.3	17.4	11.3		
		MCWB	2.8	4.3	5.8	7.5	9.6	12.4	14.2	14.6	13.0	9.8	6.5	3.7		
	10%	DB	7.9	12.1	16.1	19.0	23.0	29.1	28.8	27.5	25.2	20.8	15.0	8.7		
		MCWB	2.1	3.5	4.9	6.6	8.8	11.8	14.2	14.6	12.5	9.5	5.4	2.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	7.5	8.2	10.1	12.7	15.5	18.2	17.9	16.4	14.6	9.4	7.2		
		MCDB	9.8	15.7	18.0	21.4	26.4	24.6	23.3	24.0	23.3	19.3	18.1	11.2		
	2%	WB	5.1	5.8	7.3	8.9	11.4	14.3	17.3	16.9	15.4	12.7	8.3	5.6		
		MCDB	10.2	13.8	17.8	20.9	23.8	26.4	23.6	23.9	22.3	19.7	16.1	11.1		
	5%	WB	3.8	4.9	6.4	7.9	10.5	13.6	16.7	16.6	14.9	11.5	7.3	4.5		
		MCDB	8.6	12.7	16.2	19.4	22.6	26.7	23.3	23.8	22.1	18.3	15.6	10.0		
	10%	WB	2.7	3.9	5.6	6.9	9.6	12.8	16.1	16.1	14.1	10.3	6.3	3.3		
		MCDB	7.4	10.6	14.6	17.9	21.8	26.5	23.1	23.3	22.0	18.1	14.1	7.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	12.5	14.2	15.3	15.4	16.3	13.2	12.8	13.9	14.4	13.8	11.4		
		MCDBR	15.1	16.3	17.0	17.7	17.5	17.8	16.2	15.0	15.6	16.5	16.0	14.5		
		MCWBR	8.8	8.6	8.2	8.3	7.6	6.7	5.0	4.8	5.8	7.6	8.2	8.2		
		MCDBR	12.4	14.4	15.4	16.1	15.7	15.4	11.8	11.8	12.2	13.0	14.1	12.0		
	5% WB	MCWBR	7.8	7.8	7.6	7.6	6.7	5.8	4.0	4.2	4.7	6.7	7.5	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.236	0.240	0.263	0.271	0.282	0.296	0.344	0.344	0.300	0.269	0.244	0.236		
	taud		2.667	2.682	2.583	2.553	2.567	2.520	2.488	2.445	2.567	2.626	2.656	2.671		
	Ebn at Noon		1007	1033	1021	1017	1001	981	932	930	965	981	987	989		
	Edh at Noon		70	77	93	101	101	105	108	111	93	80	70	66		
(44) All-Sky Solar Radiation	RadAvg		3.30	4.25	5.74	7.13	7.92	8.19	6.71	6.27	5.82	4.81	3.70	2.94		
	RadStd		0.29	0.37	0.34	0.33	0.48	0.43	0.39	0.27	0.24	0.38	0.21	0.19		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	N/A	-96	+71	+28

Nomenclature: See separate page